

Report

MPOB - PALMNEX COMPREHENSIVE INLAND SOIL ANALYSIS

Estate info

- Name

Lahad Datu with block boundary
- Location

Lahad Datu, Sabah (5.10366, 118.42686)
- Current fertilizer usage

Standard MPOB Rates
- 1 block = 25 ha (basic)

25.0 ha (Block 22)



Report

Chosen Cost/Revenue Ratio: 0.30

Yield Prediction: 22.45 (t/ha/yr)

Revenue Gain: RM 6,456.00 / yr

Revenue gain formula = (Yield prediction x 25 ha x FFB price RM 600)

Element	Leaf Analysis (avg%)	Fert Recommendation (kg/palm)	Fert Recommendation (kg/ha)
N	Average per block (2.43%)	5.00	715.00
P	0.129%	0.27	38.61
K	0.85%	4.50	643.50
Mg	0.317%	0.00	0.00

Recommended Formulation:
N:7, P:3, K:30, Mg:0

Optimum Nutrient Analysis %

■ May ■ June ■ July

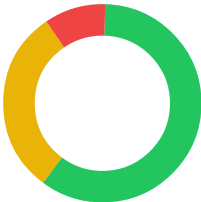


Stress-level

Stressed
10.0%

Moderate
30.0%

Healthy
60.0%



Disclaimer: This report is generated dynamically by the MPOB PalmNex system based on Sentinel-2 satellite imagery index calculations, MEDS agronomic yield matrices, and leaf nutrient analysis.

Inland Soil Analysis Input:

Characteristics

Palm age (Year)	12
Planting density (palms/ha)	148
Soil drainage class	1
Soil consistency class	1
Slope class	0.5
Root growth impedance class	0.2

Soil organic matter (%)	2
Silt (%)	18
Extractable K (cmol/kg)	0.13
Total extractable bases (cmol/kg)	1.3
Annual rainfall (mm/year)	2000

Price for F.F.B/tonne: RM 800

Cost of SOA/tonne: RM 1200

Cost of KCl/tonne: RM 1600

FFB Yield Prediction

K Fertilizer (kg/palm)

	0.0	0.5	1.0	1.5	2.0	2.5	3.0	3.5	4.0	4.5	5.0
0.0	14.38	14.85	15.23	15.49	15.68	15.80	15.86	15.88	15.86	15.81	15.73
0.5	15.22	15.76	16.22	16.54	16.80	16.98	17.11	17.18	17.21	17.21	17.18
1.0	15.90	16.52	17.05	17.44	17.78	18.01	18.20	18.33	18.42	18.47	18.49
1.5	16.32	16.99	17.59	18.04	18.43	18.72	18.96	19.14	19.27	19.36	19.42
2.0	16.63	17.37	18.03	18.54	18.99	19.33	19.63	19.85	20.04	20.17	20.27
2.5	16.81	17.60	18.31	18.87	19.37	19.75	20.09	20.35	20.57	20.74	20.88
3.0	16.92	17.76	18.53	19.14	19.68	20.11	20.50	20.80	21.06	21.26	21.44
3.5	16.96	17.85	18.66	19.31	19.89	20.36	20.78	21.11	21.41	21.64	21.84
4.0	16.97	17.90	18.76	19.44	20.07	20.57	21.03	21.39	21.72	21.98	22.21
4.5	16.95	17.92	18.81	19.53	20.18	20.71	21.20	21.59	21.94	22.22	22.48
5.0	16.91	17.91	18.84	19.59	20.28	20.83	21.35	21.76	22.14	22.45	22.73
5.5	16.86	17.89	18.84	19.62	20.33	20.91	21.45	21.89	22.29	22.61	22.91
6.0	16.80	17.85	18.84	19.64	20.38	20.98	21.54	22.00	22.42	22.76	23.08

Revenue Gain

K Fertilizer (kg/palm)

	0.0	0.5	1.0	1.5	2.0	2.5	3.0	3.5	4.0	4.5	5.0
0.0	0.00	376.00	680.00	888.00	1,040	1,136	1,184	1,200	1,184	1,144	1,080
0.5	672.00	1,104	1,472	1,728	1,936	2,080	2,184	2,240	2,264	2,264	2,240
1.0	1,216	1,712	2,136	2,448	2,720	2,904	3,056	3,160	3,232	3,272	3,288
1.5	1,552	2,088	2,568	2,928	3,240	3,472	3,664	3,808	3,912	3,984	4,032
2.0	1,800	2,392	2,920	3,328	3,688	3,960	4,200	4,376	4,528	4,632	4,712
2.5	1,944	2,576	3,144	3,592	3,992	4,296	4,568	4,776	4,952	5,088	5,200
3.0	2,032	2,704	3,320	3,808	4,240	4,584	4,896	5,136	5,344	5,504	5,648
3.5	2,064	2,776	3,424	3,944	4,408	4,784	5,120	5,384	5,624	5,808	5,968
4.0	2,072	2,816	3,504	4,048	4,552	4,952	5,320	5,608	5,872	6,080	6,264
4.5	2,056	2,832	3,544	4,120	4,640	5,064	5,456	5,768	6,048	6,272	6,480
5.0	2,024	2,824	3,568	4,168	4,720	5,160	5,576	5,904	6,208	6,456	6,680
5.5	1,984	2,808	3,568	4,192	4,760	5,224	5,656	6,008	6,328	6,584	6,824
6.0	1,936	2,776	3,568	4,208	4,800	5,280	5,728	6,096	6,432	6,704	6,960